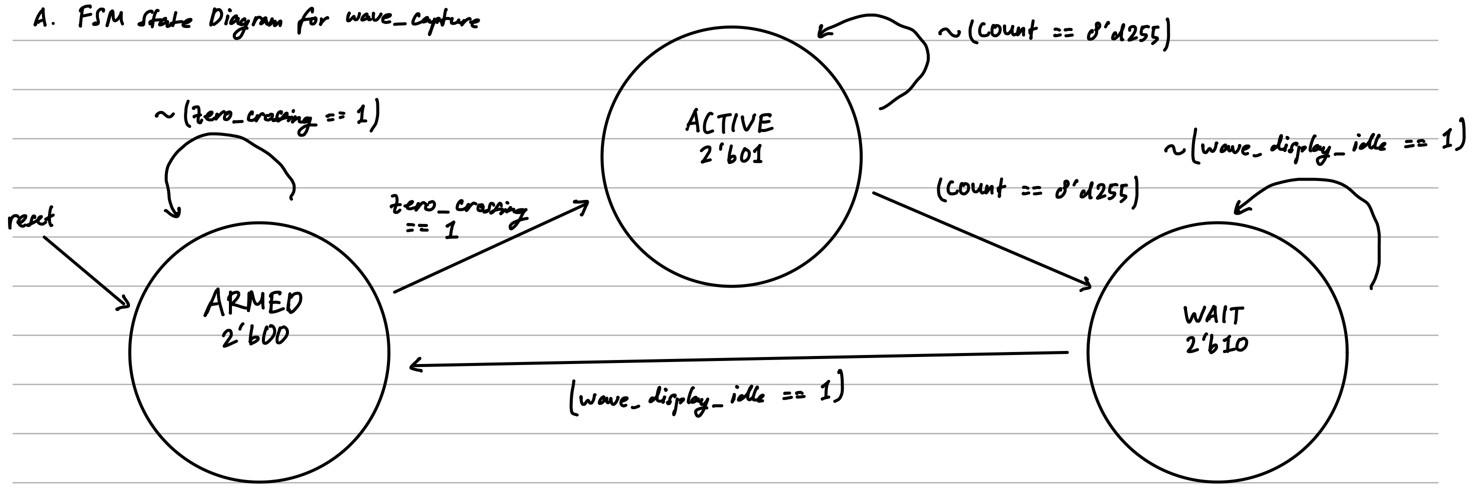
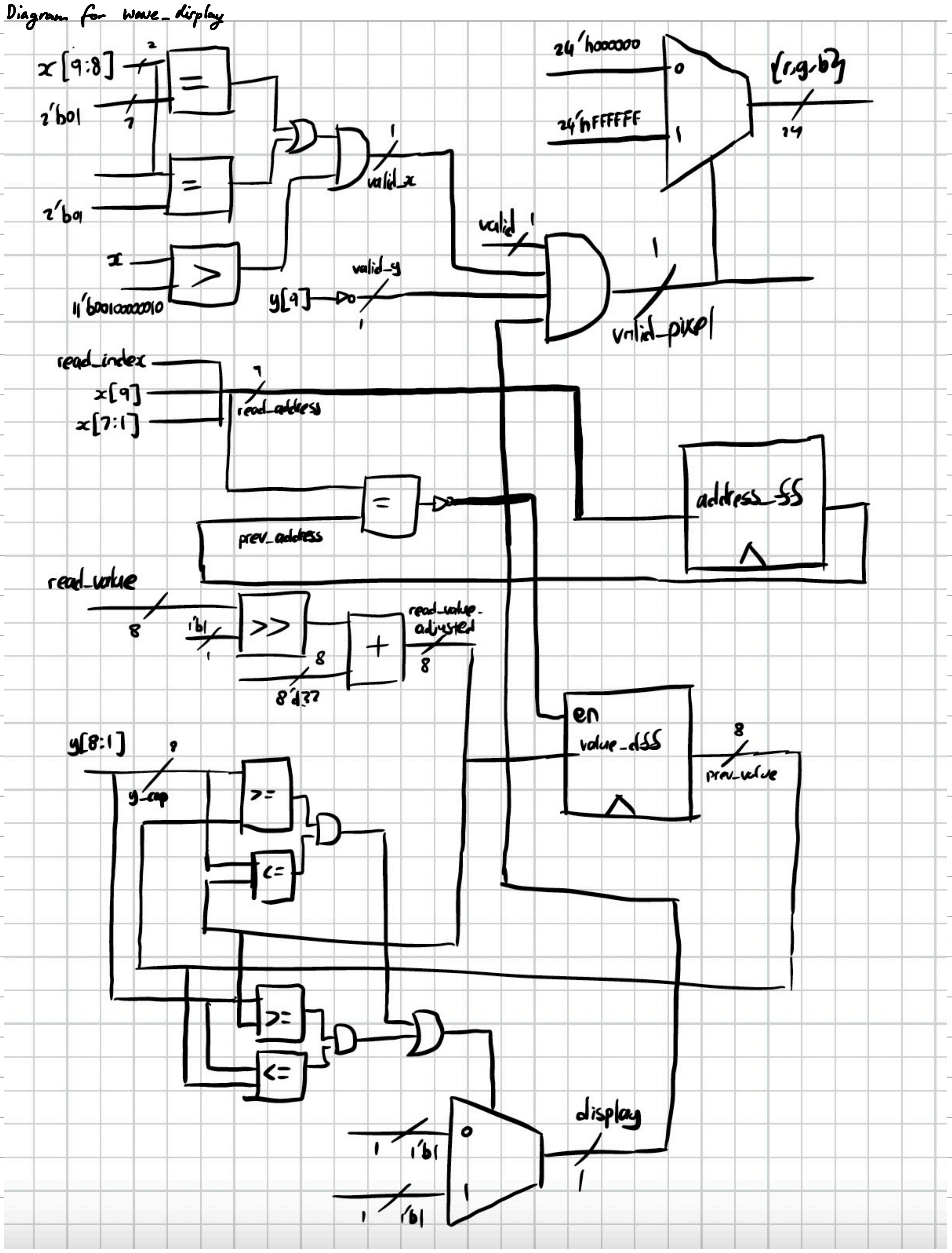
1. Diagrams

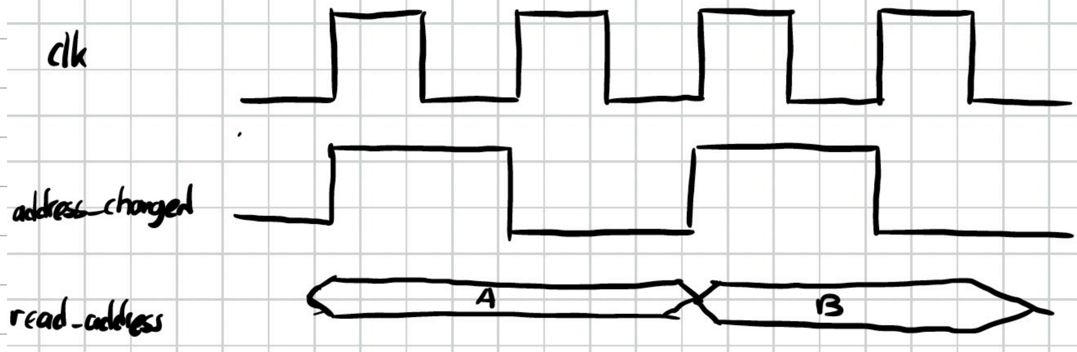
A. FSM State Diagram for wave_capture



B. Block Diagram for wave_display



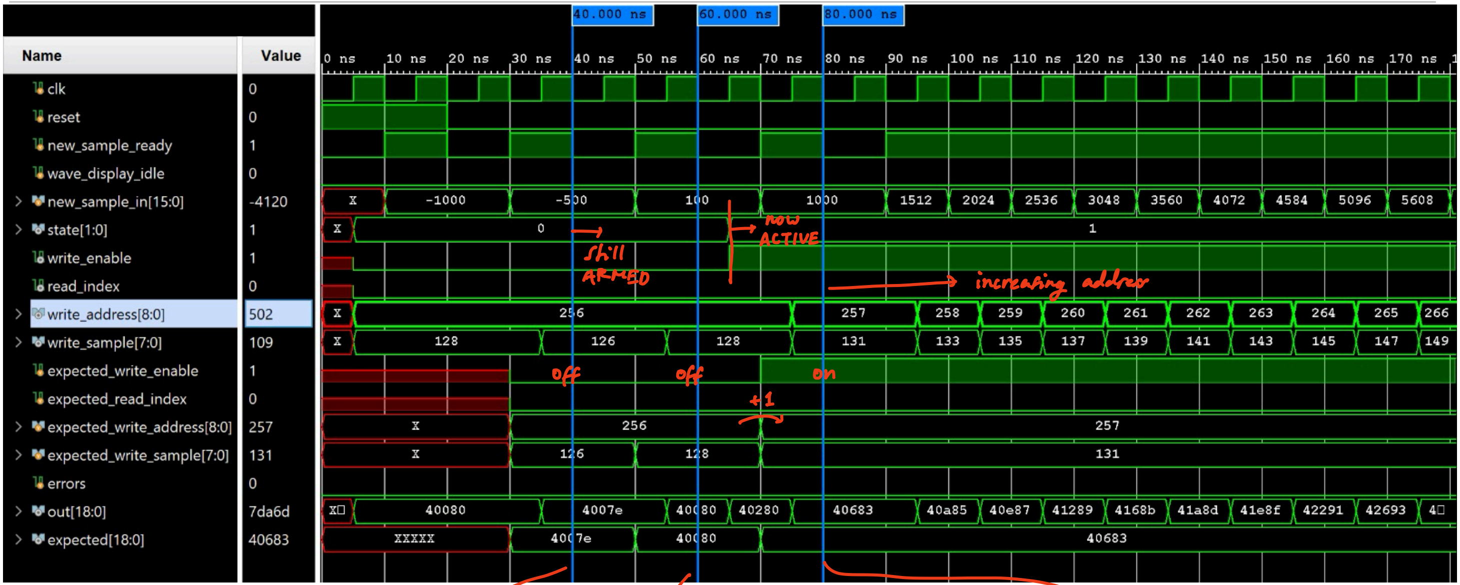
C. Timing Diagram for wave_display



When address changes from A → B, the RAM still outputs A for one clock cycle before B, showing one-cycle read latency

2. Annotated Testbenches

A. wave_capture_tb.v



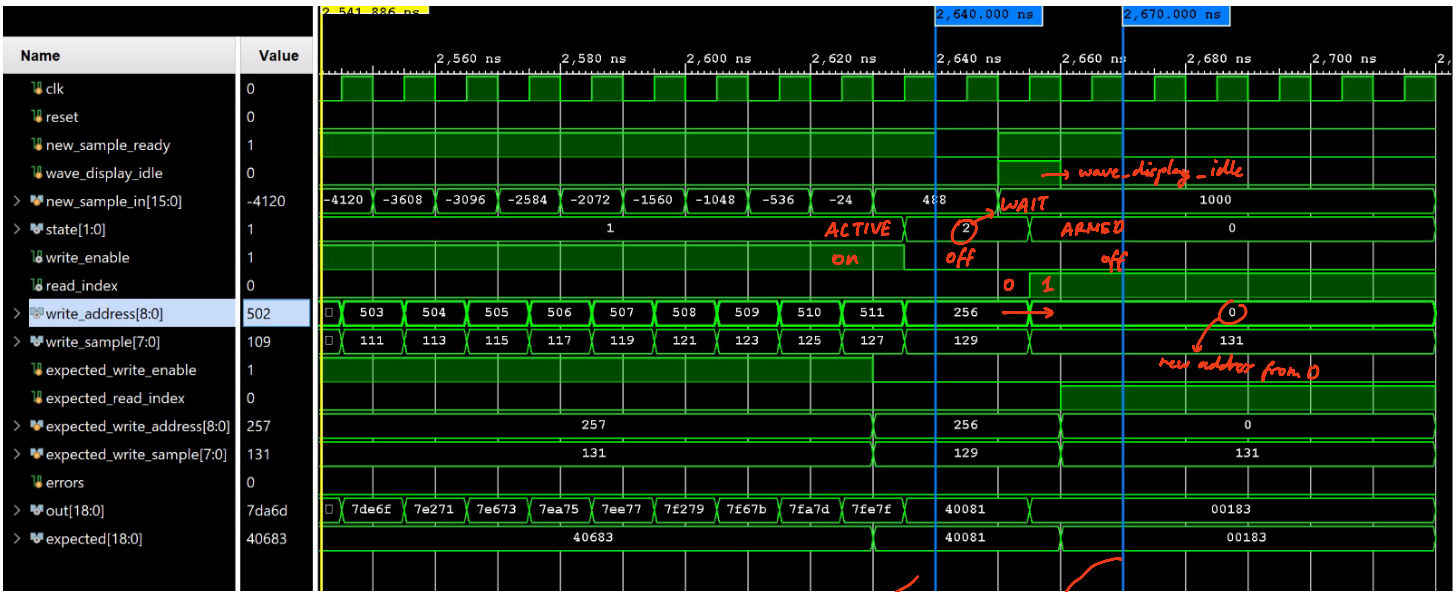
test 1 : test ARMED
sample is still negative
so there is no zero crossing

→ state is still ARMED
→ all outputs are same as initialized

test 2 : test ARMED → ACTIVE
new sample is positive
↳ there is a zero crossing

→ next state is ACTIVE
→ all outputs are same as initialized
(write_enable is off)

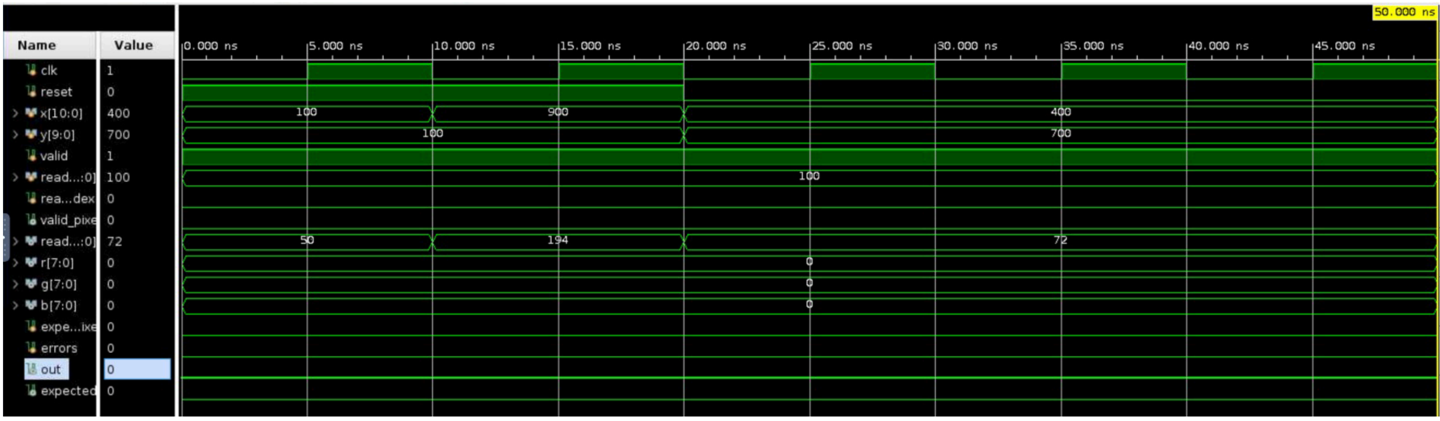
test 3 : test ACTIVE functionality
write_enable is on as expected.
address increases by 1 for every new sample.



test 4 : after 256 samples, state becomes WAIT
→ write_enable = 0
→ read_index hasn't changed yet

test 5 : wave_display_idle, WAIT → ARMED
→ write_enable = 0
→ read_index changes to 1
↳ address start from 0

B. wave_display_tb.v




```
# run 1000ns
Test1 small x  actual=0 expected=0
Test2 large x  actual=0 expected=0
Test3 bottom half y  actual=0 expected=0
PASS
$finish called at time : 50 ns : File "/home/users/berwyn/Desktop/Labs/lab5/src/sim/wave_display_tb.v" Line 79
```

3. Critical Path

Summary

Name	Path 105
Slack	2.565ns
Source	wd_top/sample_ram/RAM_reg/CLKBWRCLK (rising edge-triggered cell RAMB18E1 clocked by clk_out1_clk_wiz_0 {rise@0.000ns fall@5.000ns period=10.000ns})
Destination	U3/inst/srldly_0/srl[24].srl16_i/D (rising edge-triggered cell SRL16E clocked by clk_out2_clk_wiz_0 {rise@0.000ns fall@20.000ns period=40.000ns})
Path Group	clk_out2_clk_wiz_0
Path Type	Setup (Max at Slow Process Corner)
Requirement	10.000ns (clk_out2_clk_wiz_0 rise@40.000ns - clk_out1_clk_wiz_0 rise@30.000ns)
Data Pa... Delay	6.822ns (logic 3.359ns (49.235%) route 3.463ns (50.765%))
Logic Levels	4 (CARRY4=1 LUT4=1 LUT6=2)
Clock ... Skew	-0.363ns
Clock U...tainty	0.210ns

→ this path starts from the sample RAM in the wave_display_top module and ends in a shift (srl) register inside the provided U3 starter module.